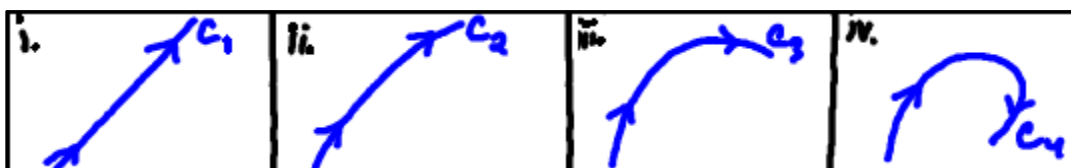


2.5 Curvature

We will now explore the notion of curvature. We begin by reviewing some terminology.

Definition: A parameterization $\mathbf{r}(t)$ is called smooth on an interval I if $\mathbf{r}'$ is continuous and $\mathbf{r}'(t) \neq \mathbf{0}$. A curve is smooth if it has a smooth parameterization. A smooth curve has no sharp corners or cusps.

Remark: For a smooth curve given by $\mathbf{r}(t)$ when the unit tangent vector, $\mathbf{T}(t)$, turns it does so continuously. When a smooth curve has sharp turns $\mathbf{T}(t)$ will turn quickly. When a curve turns quite slowly, $\mathbf{T}(t)$ will not turn quickly. Consider the pictures below for examples of this idea.



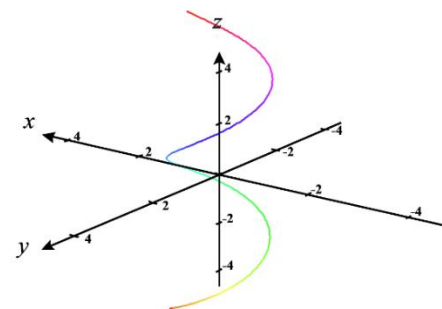
This leads to the following definition.

Definition: The curvature of a smooth curve is $\kappa = \left\| \frac{d\mathbf{T}}{ds} \right\|$.

How does the definition work?

If we can find the arc length parameterization for $\mathbf{r}(t(s))$, we can get $\mathbf{T}(s)$ and then calculate $\|\mathbf{T}'(s)\|$.

Suppose that we have the vector function $\mathbf{r}(t) = \langle \cos(t), \sin(t), t \rangle$ and what to find the curvature here.



Therefore, we have an arc length parameterization of

Now we attempt to calculate $\mathbf{T}(s) = \frac{\mathbf{r}'(s)}{\|\mathbf{r}'(s)\|}$.

Therefore,

$$\frac{d\mathbf{T}}{ds} =$$

So we say that the curvature for $\mathbf{r}(t)$ is

Theorem: The curvature of a smooth curve given by $\mathbf{r}(t)$ can be expressed as

$$1) \quad \kappa(t) = \frac{\|\mathbf{T}'(t)\|}{\|\mathbf{r}'(t)\|}.$$

OR

$$2) \quad \kappa(t) = \frac{\|\mathbf{r}'(t) \times \mathbf{r}''(t)\|}{\|\mathbf{r}'(t)\|^3}$$

Proof: Proofs of (1) and (2) are available in Blackboard.

Note: These formulas provide us with a much easier way to compute the curvature compared with computing directly from the definition.

Note: Part (b) of this theorem will tend to be more useful since it only requires that we know the vector-valued function $\mathbf{r}(t)$, we don't need to compute the tangent vector $\mathbf{T}(t)$, and we can easily change the parameter if needed.

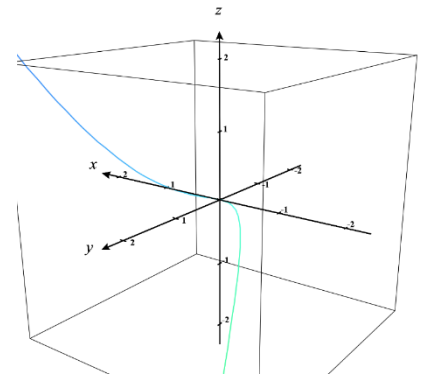
Example 1: Use each of the formulas from the theorem above to calculate the curvature of $\mathbf{r}(t) = \langle \cos(t), \sin(t), t \rangle$.

Example 2: Consider the vector function $\mathbf{r}(t) = \left\langle t, t^2, \frac{2}{3}t^3 \right\rangle$.

a) Use any approach to calculate the curvature of $\mathbf{r}(t)$.

Note: The curvature formula here depends on the value of t ! This means that the curve $\mathbf{r}(t)$ isn't equally "bendy" at all points. If we evaluate this formula at a specific value of t we can determine the curvature at that time on the curve.

Calculate the curvature when $t = 0, t = 1$, & $t = 2$.



Note: We can visually see the different amounts of "bend" on the curve itself and note that the curve appears the most "bent" when $t = 0$ (meaning we are at the point $(0,0,0)$).

Example 3: Consider the circle given by $\mathbf{r}(t) = \langle a \cos(t), a \sin(t) \rangle$ where $0 \leq t < 2\pi$ and $a > 0$. Find the curvature of this circle.

Note: Intuitively this means that if a circle has a small radius, its curvature is high. This makes sense because over a fixed amount of arc length small circles turn more abruptly than large circles.

Note: This example gives us a way to visualize the amount of “bend” that a given curvature looks like. For example, if we calculate, as in Example 1, that we have $\kappa = \frac{1}{2}$ then we know that this is the same amount of “bend” that we would see in a circle of radius 2.

Example 4: Let C be the ellipse $x^2 + 16y^2 = 16$ oriented counterclockwise. Measure the curvature at each endpoint of its major and minor axes.

Note: It is important to note here that calculating the general curvature formula would be quite messy and so it is advantageous to only calculate the curvature at the specific values of t desired here.